

Animesh Uttekar

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TECHNICAL SKILLS

Languages: Python, TypeScript, Java, JavaScript, SQL, HTML/CSS, C, C++

Frameworks and Tools: FastAPI, React, Redux, Spring Boot, Node.js, Swagger, Git, Docker, IntelliJ, VS Code

Technologies: AWS (CloudWatch, EC2, S3, Secret Manager, RDS, Lambda), MySQL, PostgreSQL

EXPERIENCE

Research Assistant (AI/Software Engineering focus), Stony Brook University | Remote, USA

Jan 2025 – Present

- Designed and developed AgentEval, a modular Python SDK to evaluate LLM-generated outputs using 20+ industry-standard reference metrics (e.g., BLEU, ROUGE, METEOR) with gold datasets, and implemented rubric-based *LLM-as-a-Judge* evaluation for no-reference tasks, enabling comprehensive and flexible model performance assessment.
- Built an automated prompt optimization pipeline that leveraged direct and indirect reward feedback from evaluation results, integrating assessments from multiple models including OpenAI GPTs, Anthropic Claude, and Hugging Face models. This pipeline continuously refined prompts to improve accuracy, coherence, and robustness across diverse tasks.
- Accelerated large-scale evaluation workflows by 3x by implementing batch parallelization with Python's ThreadPoolExecutor, significantly reducing runtime for high-volume test suites and improving experimentation throughput for researchers and engineers.

Founding Engineer, Fulloop.ai – Formerly Prepto Inc. | San Jose, California

Jun 2024 – Dec 2024

- Developed backend services in Python FastAPI and PostgreSQL, applying Pydantic and OOP design for robust, well-tested APIs.
- Engineered end-to-end AI-powered interview automation in FastAPI, integrating OpenAI API for context-driven question and follow-up generation, scaling to 1.5K+ sessions.
- Built modular search APIs using hybrid RAG (BM25 + semantic vector retrieval), significantly improving speed and ranking accuracy over boolean search across 8K+ profiles.
- Implemented S3 multipart uploads in REST APIs for high-throughput batch document ingestion.
- Added multithreading to the evaluation pipeline, increasing throughput to 50 pages/min for document evaluation.
- Created a service that uses OCR and LangChain to parse docs and run evaluations with the OpenAI API.
- Developed the landing page for [fulloop.ai](#) using TypeScript React.js, achieving an average of over 2K monthly visits within the first two months of launch.

Software Engineer, Accenture | Maharashtra, India

Jan 2021 – Jan 2023

- Developed a Synopsis Generator using Python FastAPI and React, automating the end-to-end processing of lengthy reports and reducing client report preparation time from 1.5 hours per report to just 2 minutes, significantly improving productivity and client delivery timelines.
- Built an Abbreviation Tool using FastAPI and React to automatically generate standard abbreviations from datasets, eliminating manual effort and saving over 2,100 work hours annually, while ensuring accuracy and consistency across documents.
- Implemented structured audit and error logging with detailed contextual information, integrating logs with AWS CloudWatch to enable real-time monitoring, proactive troubleshooting, and faster issue resolution.
- Automated CI/CD pipelines using AWS CloudFormation, reducing backend deployment time from 2 hours to 15 minutes, standardizing environment setup, and ensuring higher reliability and consistency across deployments.
- Designed and implemented secure file handling with AWS presigned URLs, ensuring controlled access for uploads and downloads, achieving zero security incidents in 16 months of production usage while meeting compliance and data protection requirements.

Software Engineer, S. P. Tronics | Maharashtra, India

Aug 2019 – Dec 2020

- Built real-time inverter alert APIs using Python FastAPI and seamlessly integrated them with frontend dashboards, providing instant fault detection and notifications. This system empowered engineers to quickly diagnose issues and reduce fault resolution time from several hours to under 30 minutes, improving operational efficiency and uptime.
- Developed and deployed 15+ reusable, modular UI components using React.js and TypeScript for inverter diagnostic and alert systems, standardizing the user interface, accelerating feature development, and ensuring a consistent, intuitive user experience across the platform.

PROJECTS

AI-Powered Marketing Campaign Simulator / TypeScript, React, Python, FastAPI, OpenAI, HuggingFace

- Built a **React** dashboard to configure campaigns, run simulations, and visualize funnel metrics and ROI.
- Implemented a APIs in **Python FastAPI** with Pydantic models to run a 10K-impression Monte Carlo simulation
- Modeled the funnel with vectorized **NumPy** Monte Carlo, producing an ROI fit score and tag.
- Used **gpt-oss-120b** to extract campaign metrics and generate insights, identifying key conversion bottlenecks.

EDUCATION

Stony Brook University, Stony Brook, NY

Jan 2023 – Dec 2024

Masters of Science in Computer Engineering (CGPA: 3.73/4.0)

Relevant Courses: Cloud Computing, Machine Learning, Practical AI and ML, Network Engineering